Semaev decomposition certificate: Lean decision theorem

2026-10-21; THM-SEMAEV-FALL1 Lemma 3; TASK-20260916-b55948

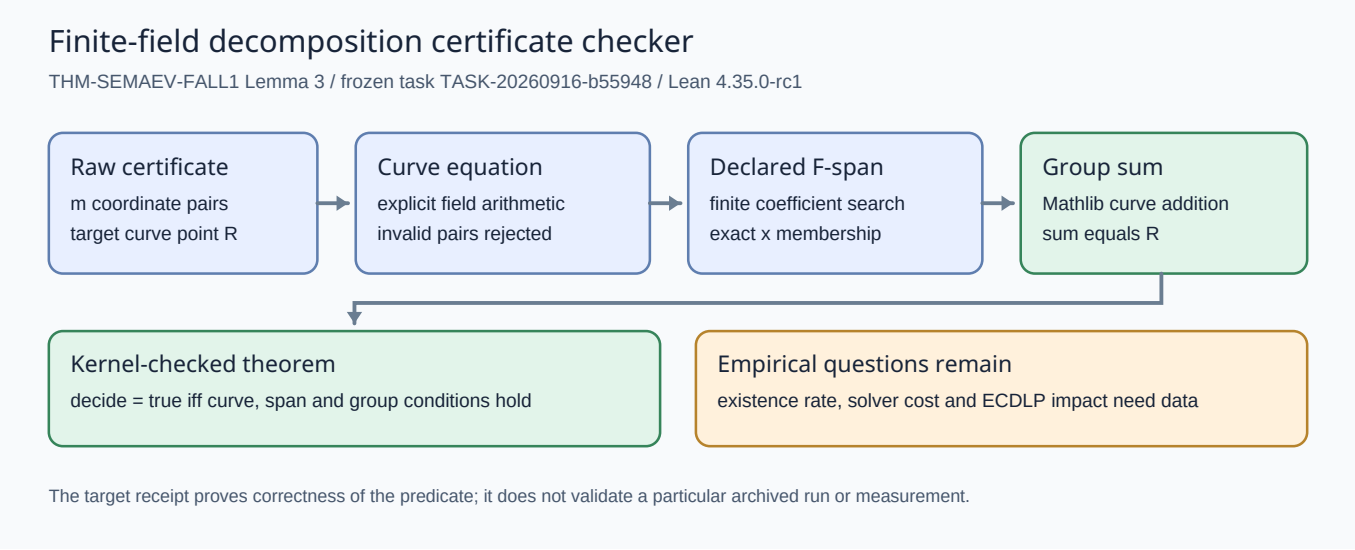
Scope. This theorem checks a proposed finite-field point decomposition. It does not assert that one exists, estimate its probability, validate a particular archived run, or bound a solver's cost.

Predicate and checked theorem

Let F be finite, K a field with decidable equality and an F-algebra structure, and $V = \mathrm{span}_F\{b_1, \ldots, b_B\} \subseteq K$. For a tuple of affine Weierstrass points $P_1, \ldots, P_m$, the checker rejects infinity, checks the curve equation and $x(P_i) \in V$, and compares the point sum with a supplied target R. Lean proves:

$$\mathrm{decide}(\mathrm{Valid}(P_1, \ldots, P_m; R)) = \texttt{true} \iff \big(\forall i,\ P_i \text{ is finite},\ E(P_i),\ x(P_i) \in V\big) \wedge \sum_i P_i = R.$$

The span decision searches the finite coefficient space F^B. A separate raw-coordinate checker rejects pairs failing the explicit curve equation, reifies valid pairs as Mathlib points, and uses the same group law. Its iff theorem connects the raw path to curve, span, and sum conditions.



Validation and limits

Check	Observed result
Pinned Lean workspace	Lake build passed 2,075 jobs on Lean 4.35.0-rc1. Project axiom audit passed 494 theorem declarations using standard axioms only.
Frozen certificate target	Machine verified; source commit and task, theorem, toolchain, and manifest hashes bound in the receipt. Independent semantic review remains pending.
Complete target corpus	64 machine verified; zero failed; zero infrastructure failures; two theorem files absent (NCP reachability and commutator-ideal refutation).

The first target attempt failed the verifier's named-theorem audit because it exported only a `Decidable` instance. The final source retains that instance and proves the named `decide = true` equivalence. No numerical dashboard or performance graph changed: no new timing datum was produced. Membership search can be expensive for large F^B; this proof gives no runtime bound or decomposition-rate estimate.